# ASSIGNMENT 9.1

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## Problem 1: Documentation styles for `find_max`

**Given function (Python):**

```
def find_max(numbers):
    return max(numbers)
```

**Task:**

- Write documentation for the function in all three formats:
  - (a) Docstring
  - (b) Inline comments
  - (c) Google-style documentation
- Critically compare the three approaches. Discuss advantages, disadvantages, and suitable use cases of each style.
- Recommend which documentation style is most effective for a mathematical utilities library and justify your answer.

**(a) Docstring**

1 Docstring.py

Assignment 9.1 > 1 Docstring.py > ...

```python
# Docstring Documentation
def find_max(numbers):
    """
    Find the maximum number in a list.
    Parameters:
    numbers: A list of numbers.
    Returns:
    The maximum number in the list.
    """
    return max(numbers)
```

**(b) Inline comments**

1 Inline.py

Assignment 9.1 > 1 Inline.py > ...

```python
#Inline comments
def find_max(numbers):
    # Use the built-in max function to find the maximum number in the list
    return max(numbers) # Return the maximum number found in the list

```

**(c) Google-style documentation**

```python
# Google-style docstring
def find_max(numbers: list) -> float:
    """
    Find the maximum number in a list.

    Args:
        numbers(list): A list of numbers.

    Returns:
        The maximum number in the list. (float)
    Example:
        >>> find_max([1, 2, 3, 4, 5])
        5
    """

    return max(numbers)
```

### Critical comparison

| Aspect | Docstring | Inline Comments | Google-style |
|---|---|---|---|
| Readability | Good | Minimal | Excellent |
| IDE Support | Native | None | Native (very good) |
| Formality | Standard | Casual | Professional |
| Maintenance | Easy | Tedious | Structured |
| Type hints / parameter clarity | Optional | Unclear | Explicit |
| Examples | Rare | Never | Common |

| Format | Advantages | Disadvantages |
|---|---|---|
| Docstring | Works with IDE tooltips and help(), simple to add | Can be less structured if not written carefully |
| Inline comments | Explains specific lines of logic right where they occur | Can clutter code and is not used by documentation tools |
| Google-style | Professional, structured, supports types and examples well | More verbose and takes more time to write |

**Recommendation (for a mathematical utilities library):**

Google-style documentation is the most effective because it is clear, structured, and consistent across many functions. It also scales well and supports automated documentation generation.

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## Problem 2: Documentation styles for `login`

**Given function (Python):**

```
def login(user, password, credentials):
    return credentials.get(user) == password
```

**Task:**

1. Write documentation in all three formats.
2. Critically compare the approaches.
3. Recommend which style would be most helpful for new developers onboarding a project, and justify your choice.

**Outputs**

```python
# Docstring Type Documentation
def login(user, password, credentials):
    """
    Logs in a user by checking if the provided password matches the stored credentials.
    Parameters:
    user: The username of the user trying to log in.
    password: The password provided by the user.
    credentials: A dictionary containing usernames as keys and their corresponding passwords as values.
    Returns:
    True if the login is successful (i.e., the password matches the stored credentials), False otherwise.
    """
    return credentials.get(user) == password

```

```python
# Google Style Docstring
def login(user: str, password: str, credentials: dict) -> bool:
    """
    Logs in a user by checking if the provided password matches the stored credentials.

    Args:
        user (str): The username of the user trying to log in.
        password (str): The password provided by the user.
        credentials (dict): A dictionary containing usernames as keys and their corresponding passwords
        as values.

    Returns:
        bool: True if the login is successful (i.e., the password matches the stored credentials),
        False otherwise.
    Example:
        >>> credentials = {'alice': 'password123', 'bob': 'securepass'}
        >>> login('alice', 'password123', credentials)
        True
    Raises ValueError: If the user is not found in the credentials.
    """

    if user not in credentials:
        raise ValueError("User not found in credentials.")
    return credentials[user] == password
```

```python
# Inline Comment
def login(user, password, credentials):
    return credentials.get(user) == password # This line checks if the password provided by the
#user matches the password stored in the credentials dictionary for that user
# ---------------------------------------------------------------------------------------------
# The function 'login' checks if the provided password matches the stored credentials for a given user.
# It uses the 'get' method of the dictionary to retrieve the password associated with the user,
# returning True if it matches the provided password and False otherwise.
# ---------------------------------------------------------------------------------------------
```

### Comparison

| Point | Inline | Docstring | Google Style |
|---|---|---|---|
| Clear for beginners | No | Yes | Yes (very clear) |
| Structured | No | Somewhat | Yes |
| Professional | No | Medium | Yes |
| Best for large projects | No | Okay | Yes |
| Best for onboarding | No | Good | Best |

### Recommendation (for onboarding):

Google-style documentation is the most helpful because new developers need clarity, structure, and consistency. It clearly explains inputs, outputs, and edge cases, and it supports automated documentation generation.

## Problem 3: Calculator (Automatic Documentation Generation)

**Task:** Design a Python module named `calculator.py` and demonstrate automatic documentation generation.

**Instructions:**

1. Create `calculator.py` with the following functions (with appropriate docstrings):
   - `add(a, b)` — returns the sum of two numbers
   - `subtract(a, b)` — returns the difference of two numbers
   - `multiply(a, b)` — returns the product of two numbers
   - `divide(a, b)` — returns the quotient of two numbers
2. Display the module documentation in the terminal using Python documentation tools.
3. Generate and export the module documentation in HTML format using `pydoc`, and open the generated HTML file in a web browser to verify the output.

**Outputs**

Calculator.py

Assignment 9.1 > Calculator.py > multiply

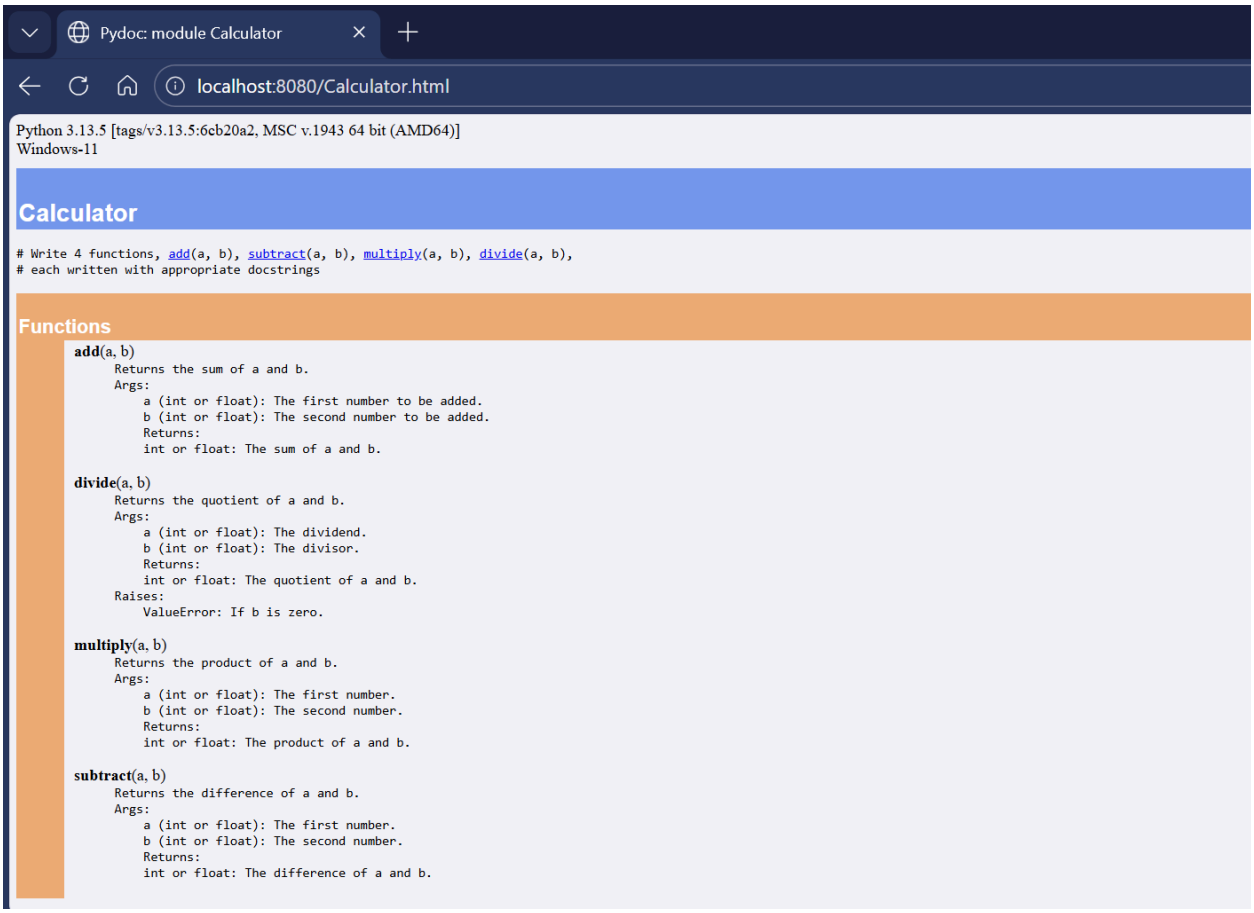
```python
# Write 4 functions, add(a, b), subtract(a, b), multiply(a, b), divide(a, b),
# each written with appropriate docstrings
def add(a, b):
    """
    Returns the sum of a and b.
    Args:
        a (int or float): The first number to be added.
        b (int or float): The second number to be added.
        Returns:
        int or float: The sum of a and b.
    """
    return a + b

def subtract(a, b):
    """
    Returns the difference of a and b.
    Args:
        a (int or float): The first number.
        b (int or float): The second number.
        Returns:
        int or float: The difference of a and b.
    """
    return a - b
```

```python
def multiply(a, b):
    """
    Returns the product of a and b.
    Args:
        a (int or float): The first number.
        b (int or float): The second number.
        Returns:
        int or float: The product of a and b.
    """
    return a * b

def divide(a, b):
    """
    Returns the quotient of a and b.
    Args:
        a (int or float): The dividend.
        b (int or float): The divisor.
        Returns:
        int or float: The quotient of a and b.
    Raises:
        ValueError: If b is zero.
    """
    if b == 0:
        raise ValueError("Cannot divide by zero.")
    return a / b

print(add.__doc__)
print(subtract.__doc__)
print(multiply.__doc__)
print(divide.__doc__)
```

```
PS Z:\AIAC\Assignment 9.1> python -m pydoc .\Calculator.py
```

```
Returns the sum of a and b.
Args:
    a (int or float): The first number to be added.
    b (int or float): The second number to be added.
    Returns:
    int or float: The sum of a and b.


Returns the difference of a and b.
Args:
    a (int or float): The first number.
    b (int or float): The second number.
    Returns:
    int or float: The difference of a and b.


Returns the product of a and b.
Args:
    a (int or float): The first number.
    b (int or float): The second number.
    Returns:
    int or float: The product of a and b.


Returns the quotient of a and b.
Args:
    a (int or float): The dividend.
    b (int or float): The divisor.
    Returns:
    int or float: The quotient of a and b.
Raises:
    ValueError: If b is zero.
```

```
PS Z:\AIAC\Assignment 9.1> python -m pydoc -w .\Calculator.py

  Returns the sum of a and b.
  Args:
      a (int or float): The first number to be added.
      b (int or float): The second number to be added.
      Returns:
      int or float: The sum of a and b.


  Returns the difference of a and b.
  Args:
      a (int or float): The first number.
      b (int or float): The second number.
      Returns:
      int or float: The difference of a and b.


  Returns the product of a and b.
  Args:
      a (int or float): The first number.
      b (int or float): The second number.
      Returns:
      int or float: The product of a and b.


  Returns the quotient of a and b.
  Args:
      a (int or float): The dividend.
      b (int or float): The divisor.
      Returns:
      int or float: The quotient of a and b.
  Raises:
      ValueError: If b is zero.

  wrote Calculator.html
```

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## Problem 4: Conversion Utilities Module

### Task:

1. Write a module named `conversion.py` with functions:
   - `decimal_to_binary(n)`
   - `binary_to_decimal(b)`
   - `decimal_to_hexadecimal(n)`
2. Use Copilot for auto-generating docstrings.
3. Generate documentation in the terminal.
4. Export the documentation in HTML format and open it in a browser.

### Outputs

conversion.py

Assignment 9.1 > conversion.py > ...

```python
# Write 3 functions decimal_to_binary(n), binary_to_decimal(s), and decimal_to_hexadecimal(n),
# each written with appropriate docstrings
def decimal_to_binary(n):
    """
    Converts a decimal number to its binary representation.
    Args:
        n (int): The decimal number to be converted.
    Returns:
        str: The binary representation of the decimal number.
    """
    if n < 0:
        raise ValueError("Input must be a non-negative integer.")
    return bin(n)[2:]
def binary_to_decimal(s):
    """
    Converts a binary string to its decimal representation.
    Args:
        s (str): The binary string to be converted.
    Returns:
        int: The decimal representation of the binary string.
    """
    if not all(char in '01' for char in s):
        raise ValueError("Input must be a valid binary string.")
    return int(s, 2)
def decimal_to_hexadecimal(n):
    """
    Converts a decimal number to its hexadecimal representation.
    Args:
        n (int): The decimal number to be converted.
    Returns:
        str: The hexadecimal representation of the decimal number.
    """
    if n < 0:
        raise ValueError("Input must be a non-negative integer.")
    return hex(n)[2:].upper()
```

```
PS Z:\AIAC\Assignment 9.1> python -m pydoc .\conversion.py

 Converts a decimal number to its binary representation.
 Args:
     n (int): The decimal number to be converted.
 Returns:
     str: The binary representation of the decimal number.


 Converts a binary string to its decimal representation.
 Args:
     s (str): The binary string to be converted.
 Returns:
     int: The decimal representation of the binary string.


 Converts a decimal number to its hexadecimal representation.
 Args:
     n (int): The decimal number to be converted.
 Returns:
     str: The hexadecimal representation of the decimal number.

 Help on module conversion:

 NAME
     conversion

 DESCRIPTION
     # Write 3 functions decimal_to_binary(n), binary_to_decimal(s), and decimal_to_hexadecimal(n),
     # each written with appropriate docstrings

 FUNCTIONS
     binary_to_decimal(s)
         Converts a binary string to its decimal representation.
         Args:
             s (str): The binary string to be converted.
         Returns:
             int: The decimal representation of the binary string.
```

```
     decimal_to_binary(n)
         Converts a decimal number to its binary representation.
         Args:
             n (int): The decimal number to be converted.
         Returns:
             str: The binary representation of the decimal number.

     decimal_to_hexadecimal(n)
         Converts a decimal number to its hexadecimal representation.
         Args:
             n (int): The decimal number to be converted.
         Returns:
             str: The hexadecimal representation of the decimal number.

 FILE
     z:\aiac\assignment 9.1\conversion.py
```

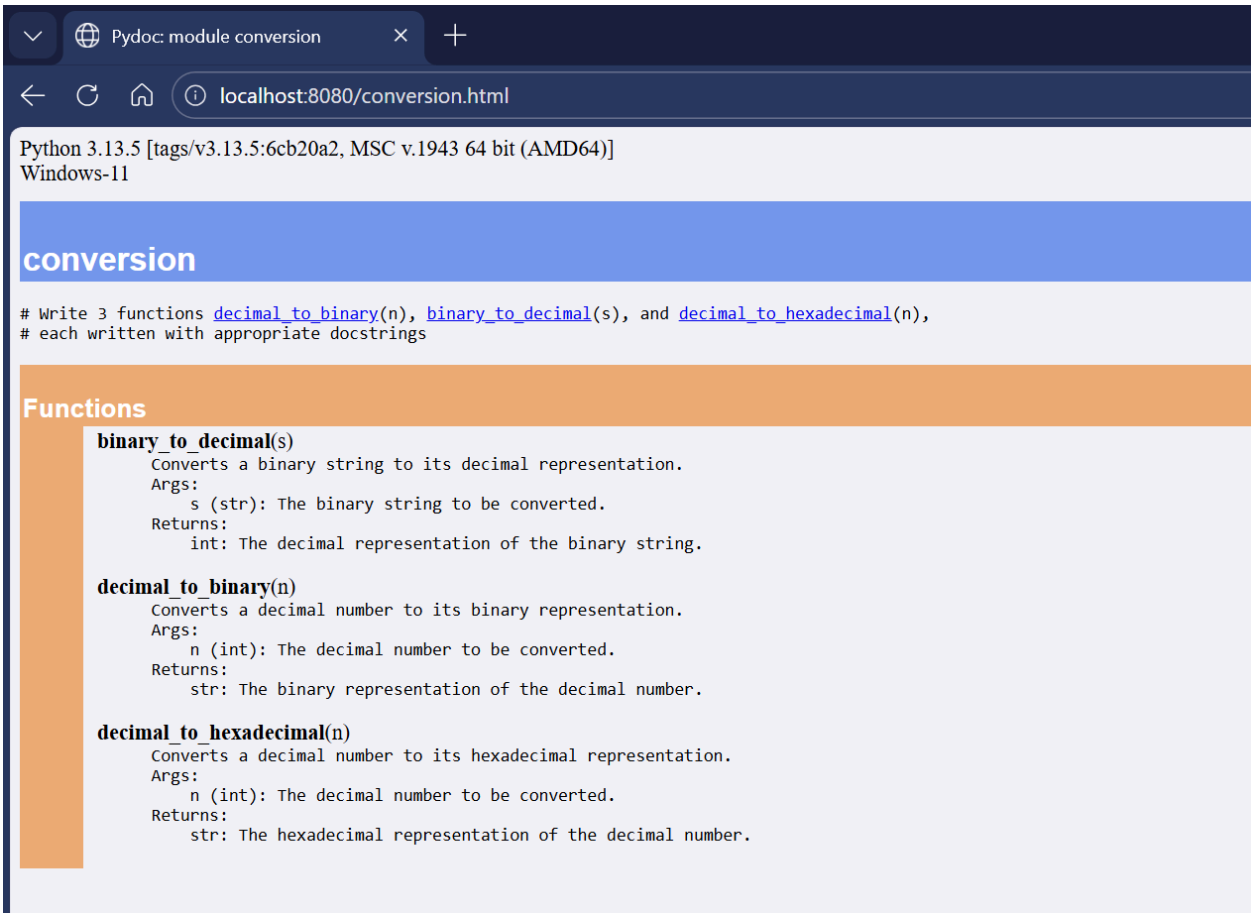
```
PS Z:\AIAC\Assignment 9.1> python -m pydoc -w .\conversion.py

 Converts a decimal number to its binary representation.
 Args:
     n (int): The decimal number to be converted.
 Returns:
     str: The binary representation of the decimal number.


 Converts a binary string to its decimal representation.
 Args:
     s (str): The binary string to be converted.
 Returns:
     int: The decimal representation of the binary string.


 Converts a decimal number to its hexadecimal representation.
 Args:
     n (int): The decimal number to be converted.
 Returns:
     str: The hexadecimal representation of the decimal number.

 wrote conversion.html
```

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## Problem 5: Course Management Module

### Task:

1. Create a module `course.py` with functions:
   - `add_course(course_id, name, credits)`
   - `remove_course(course_id)`
   - `get_course(course_id)`
2. Add docstrings with Copilot.
3. Generate documentation in the terminal.
4. Export the documentation in HTML format and open it in a browser.

### Outputs

course.py

Assignment 9.1 > course.py > remove_course

```python
# Create 3 functions add_course(course_id, name, credits), remove_course(course_id),
# get_course(course_id), with proper docstrings.
def add_course(course_id, name, credits):
    """
    Adds a course to the course list.

    Parameters:
    course_id (str): The unique identifier for the course.
    name (str): The name of the course.
    credits (int): The number of credits for the course.

    Returns:
    dict: A dictionary representing the added course.
    """
    course = {
        'course_id': course_id,
        'name': name,
        'credits': credits
    }
    return course

```

```python
def remove_course(course_id, course_list):
    """
    Removes a course from the course list.

    Parameters:
    course_id (str): The unique identifier for the course to be removed.
    course_list (list): The list of courses from which to remove the course.

    Returns:
    bool: True if the course was successfully removed, False otherwise.
    """
    for course in course_list:
        if course['course_id'] == course_id:
            course_list.remove(course)
            return True
    return False

def get_course(course_id, course_list):
    """
    Retrieves a course from the course list.

    Parameters:
    course_id (str): The unique identifier for the course to be retrieved.
    course_list (list): The list of courses from which to retrieve the course.

    Returns:
    dict: A dictionary representing the retrieved course, or None if the course is not found.
    """
    for course in course_list:
        if course['course_id'] == course_id:
            return course
    return None
```

```
PS Z:\AIAC\Assignment 9.1> python -m pydoc .\course.py

 Adds a course to the course list.

 Parameters:
 course_id (str): The unique identifier for the course.
 name (str): The name of the course.
 credits (int): The number of credits for the course.

 Returns:
 dict: A dictionary representing the added course.


 Removes a course from the course list.

 Parameters:
 course_id (str): The unique identifier for the course to be removed.
 course_list (list): The list of courses from which to remove the course.

 Returns:
 bool: True if the course was successfully removed, False otherwise.


 Retrieves a course from the course list.

 Parameters:
 course_id (str): The unique identifier for the course to be retrieved.
 course_list (list): The list of courses from which to retrieve the course.
```

```
PS Z:\AIAC\Assignment 9.1> python -m pydoc -w .\course.py

  Adds a course to the course list.

  Parameters:
  course_id (str): The unique identifier for the course.
  name (str): The name of the course.
  credits (int): The number of credits for the course.

  Returns:
  dict: A dictionary representing the added course.


  Removes a course from the course list.

  Parameters:
  course_id (str): The unique identifier for the course to be removed.
  course_list (list): The list of courses from which to remove the course.

  Returns:
  bool: True if the course was successfully removed, False otherwise.


  Retrieves a course from the course list.

  Parameters:
  course_id (str): The unique identifier for the course to be retrieved.
  course_list (list): The list of courses from which to retrieve the course.

  Returns:
  dict: A dictionary representing the retrieved course, or None if the course is not found.

  wrote course.html
```

Pydoc: module course

localhost:8080/course.html

# course

```
# Create 3 functions add_course(course_id, name, credits), remove_course(course_id),
# get_course(course_id), with proper docstrings.
```

## Functions

**add_course**(course_id, name, credits)

```
Adds a course to the course list.

Parameters:
course_id (str): The unique identifier for the course.
name (str): The name of the course.
credits (int): The number of credits for the course.

Returns:
dict: A dictionary representing the added course.
```

**get_course**(course_id, course_list)

```
Retrieves a course from the course list.

Parameters:
course_id (str): The unique identifier for the course to be retrieved.
course_list (list): The list of courses from which to retrieve the course.

Returns:
dict: A dictionary representing the retrieved course, or None if the course is not found.
```

**remove_course**(course_id, course_list)

```
Removes a course from the course list.

Parameters:
course_id (str): The unique identifier for the course to be removed.
course_list (list): The list of courses from which to remove the course.

Returns:
bool: True if the course was successfully removed, False otherwise.
```